

SIR Model: Chickenpox

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Objective

To analyze the dynamics of the Susceptibles-Infectious-Recovered (SIR) epidemic model as applied to chickenpox.

Remove all objects from workspace.

```
remove (list = objects() )
```

Load add-on packages - deSolve - contains lsoda function - differential equation solver.

```
library (deSolve)
```

```
##  
## Attaching package: 'deSolve'  
##  
## The following object is masked from 'package:graphics':  
##  
##      matplot
```

Function to compute derivatives of the differential equations.

```
sir_model = function (current_timepoint, state_values, parameters)  
{  
  # create state variables (local variables)  
  S = state_values [1]      # susceptibles  
  I = state_values [2]      # infectious  
  R = state_values [3]      # recovered  
  
  with (  
    as.list (parameters),    # variable names within parameters can be used  
    {  
      # compute derivatives  
      dS = (-beta * S * I)  
      dI = ( beta * S * I) - (gamma * I)  
      dR = (gamma * I)  
  
      # combine results  
      results = c (dS, dI, dR)  
      list (results)  
    }  
  )  
}
```

Parameters